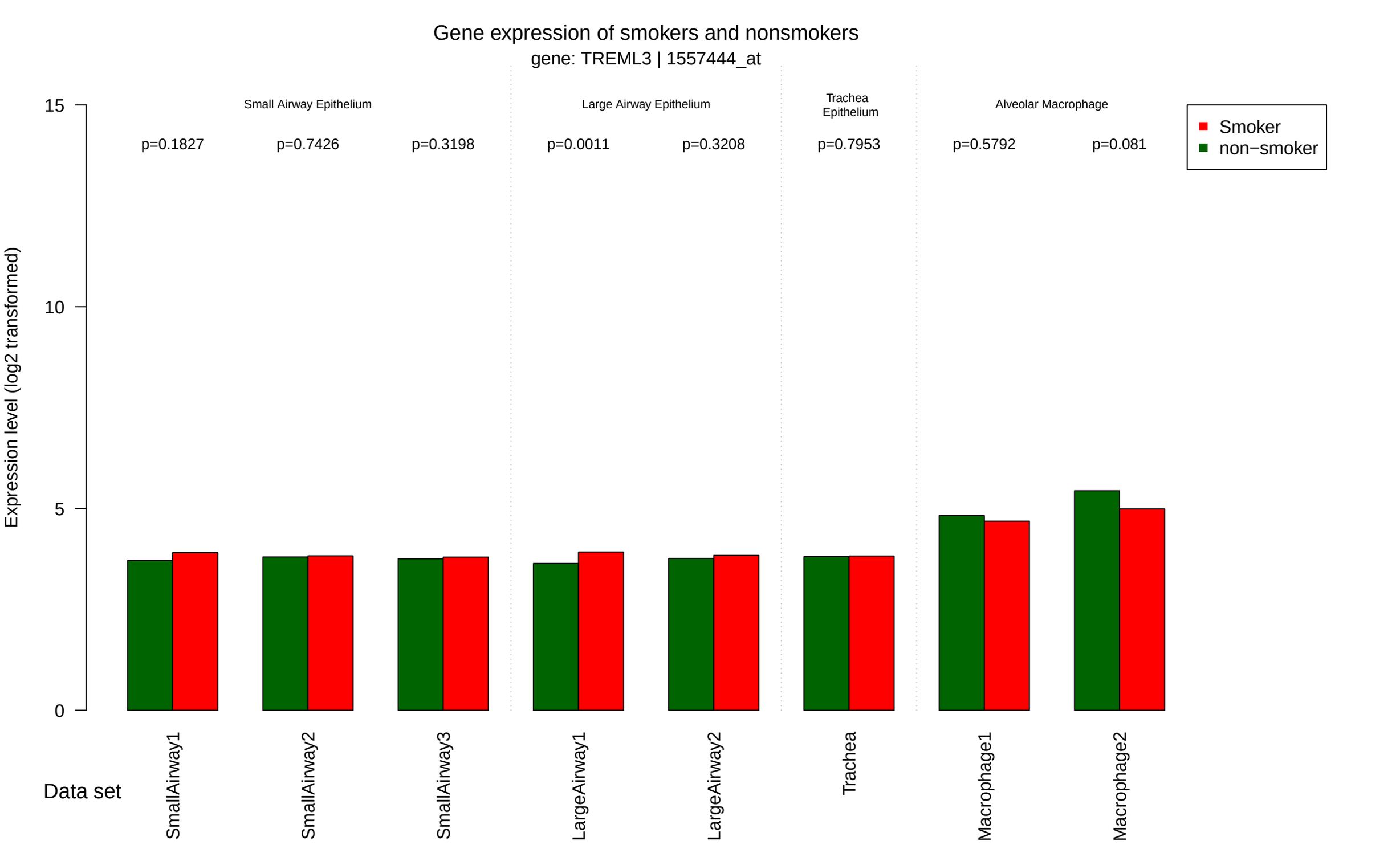
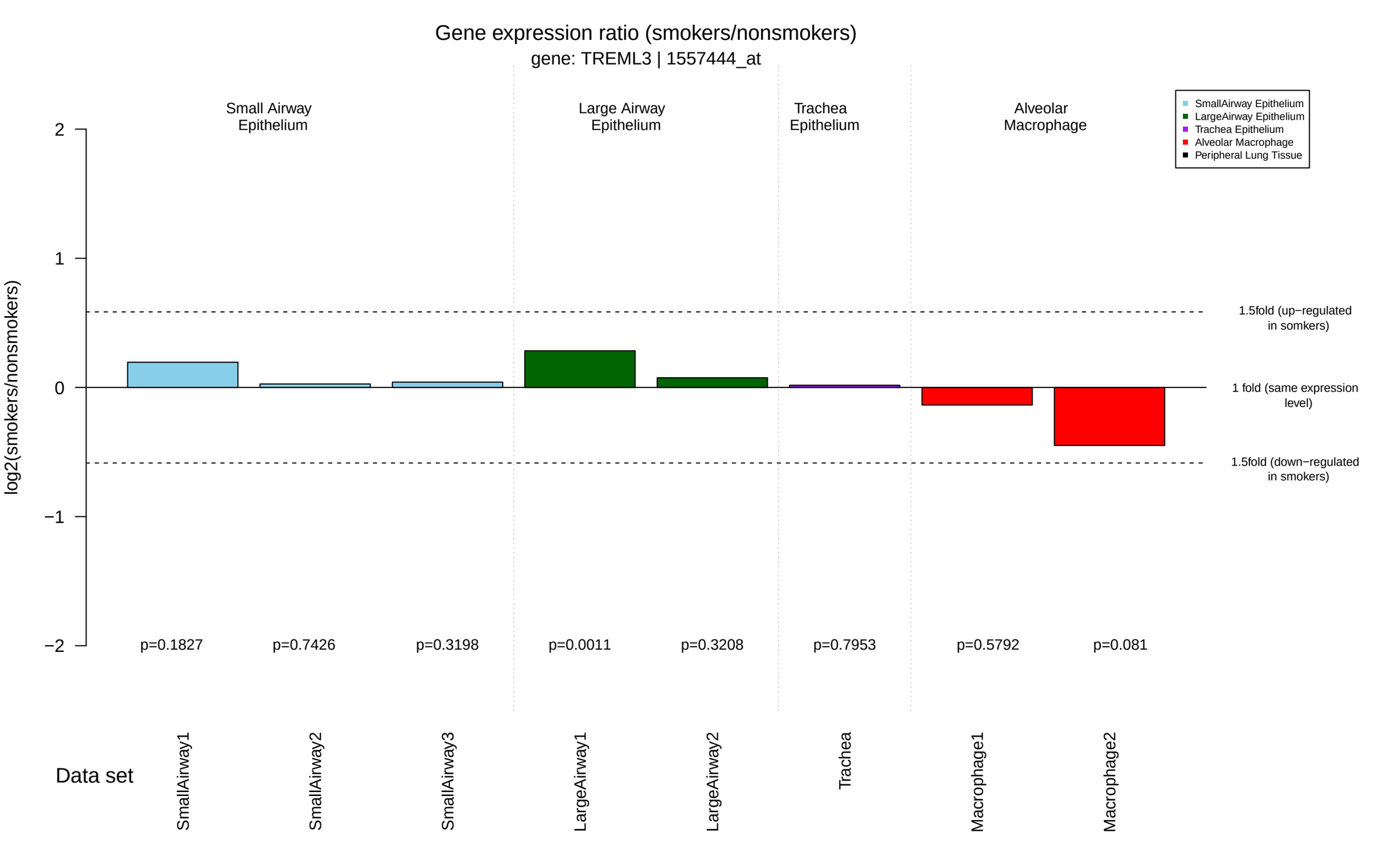
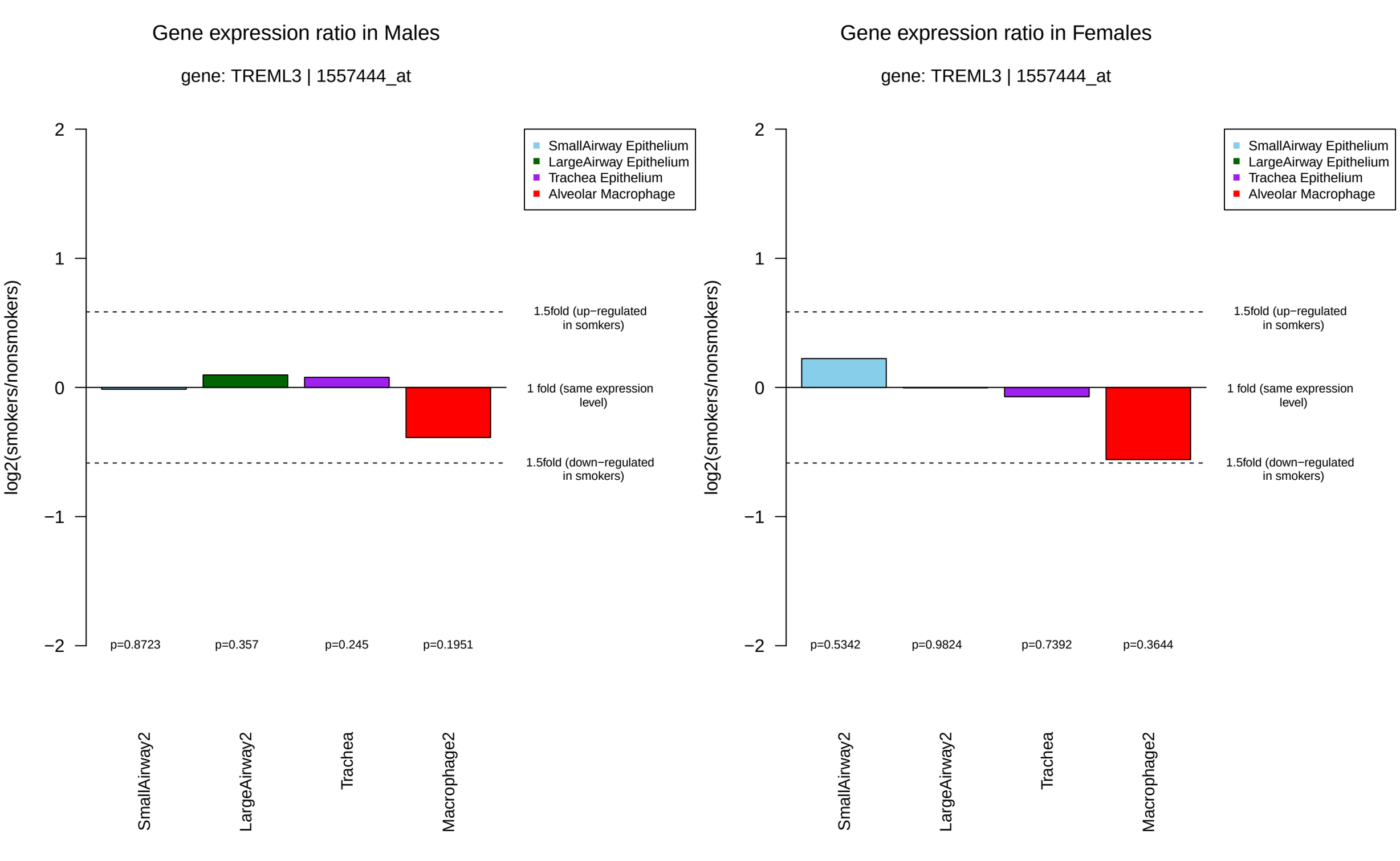
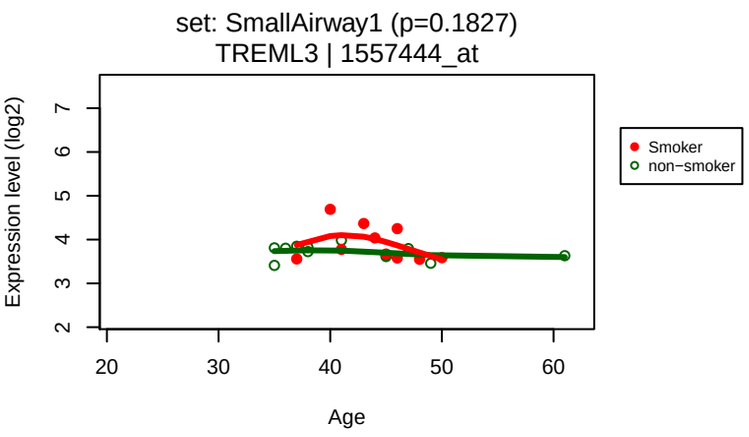
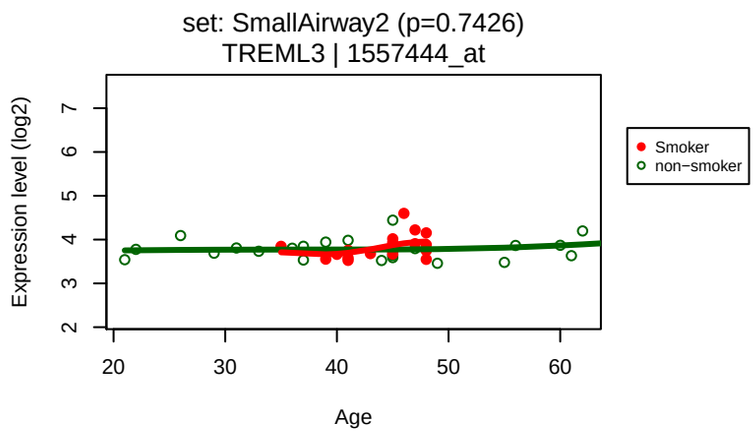
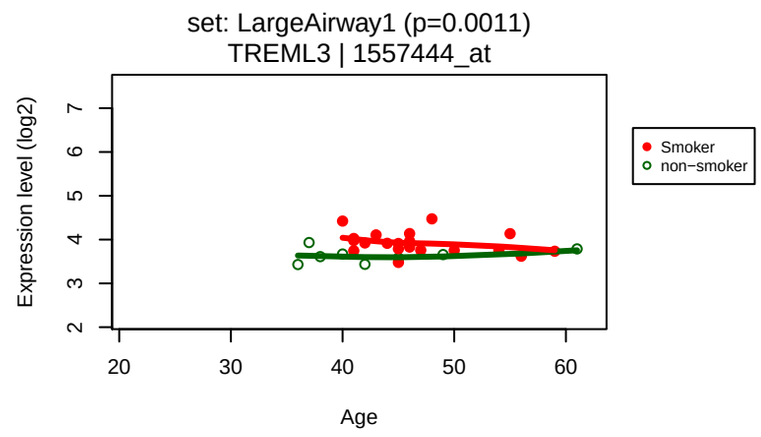

Gene expression of smokers and nonsmokers
gene: TREML3 | 1557444_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1827
p=0.7426
p=0.3198
p=0.0011
p=0.3208
p=0.7953
p=0.5792
p=0.081
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

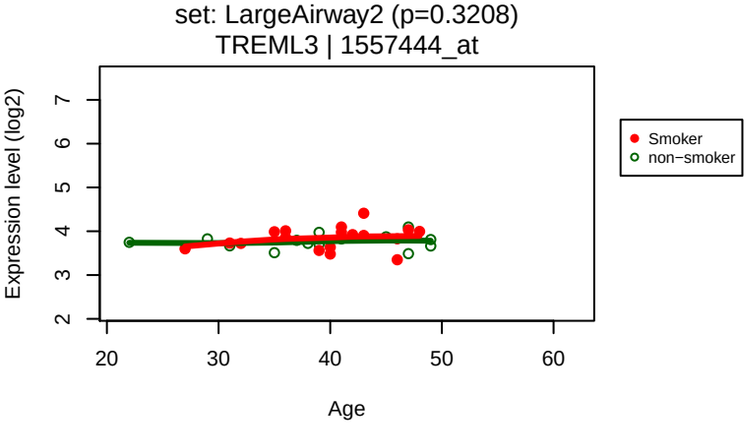

Gene expression ratio (smokers/nonsmokers)
gene: TREML3 | 1557444_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1827
p=0.7426
p=0.3198
p=0.0011
p=0.3208
p=0.7953
p=0.5792
p=0.081
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

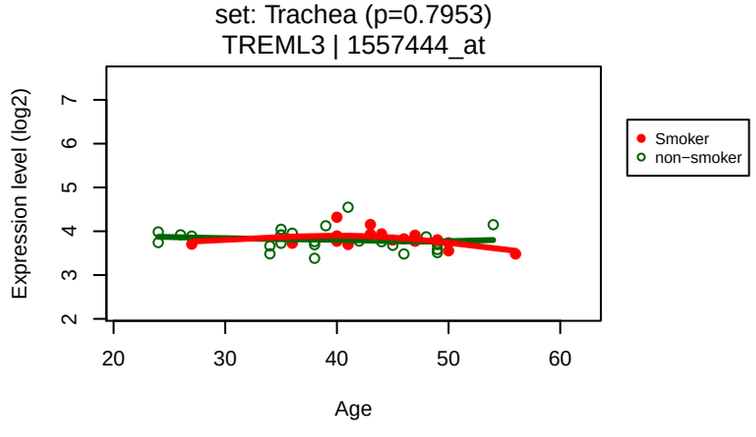

Gene expression ratio in Males
gene: TREML3 | 1557444_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8723
p=0.357
p=0.245
p=0.1951
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TREML3 | 1557444_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5342
p=0.9824
p=0.7392
p=0.3644
SmallAirway2
LargeAirway2
Trachea
Macrophage2

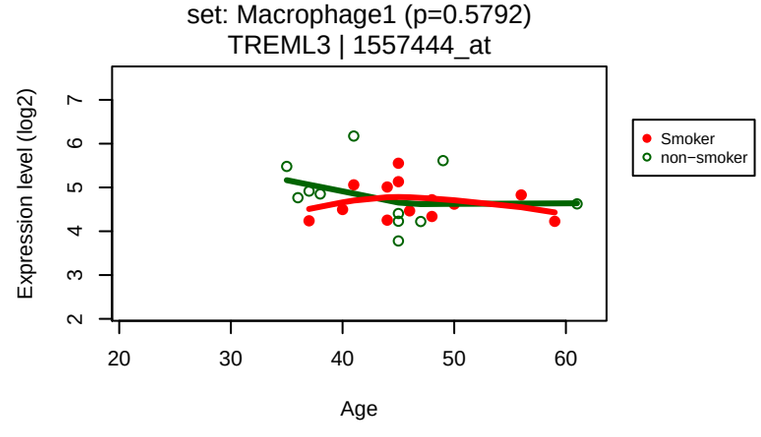

set: SmallAirway1 (p=0.1827)
TREML3 | 1557444_at
Expression level (log2)
Age
Smoker
non−smoker

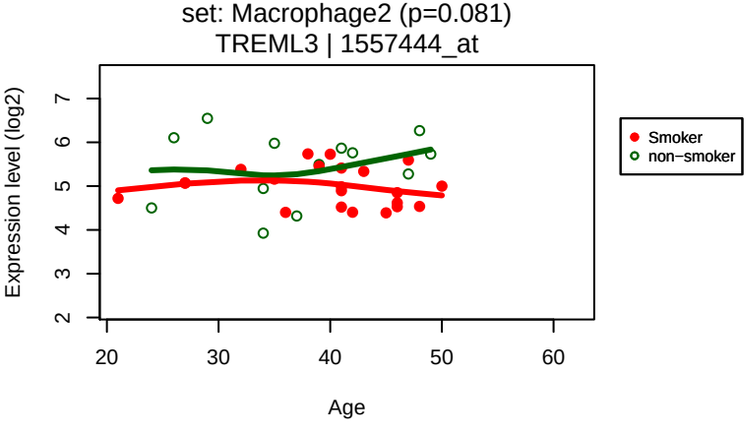

set: SmallAirway2 (p=0.7426)
TREML3 | 1557444_at
Expression level (log2)
Age
Smoker
non−smoker

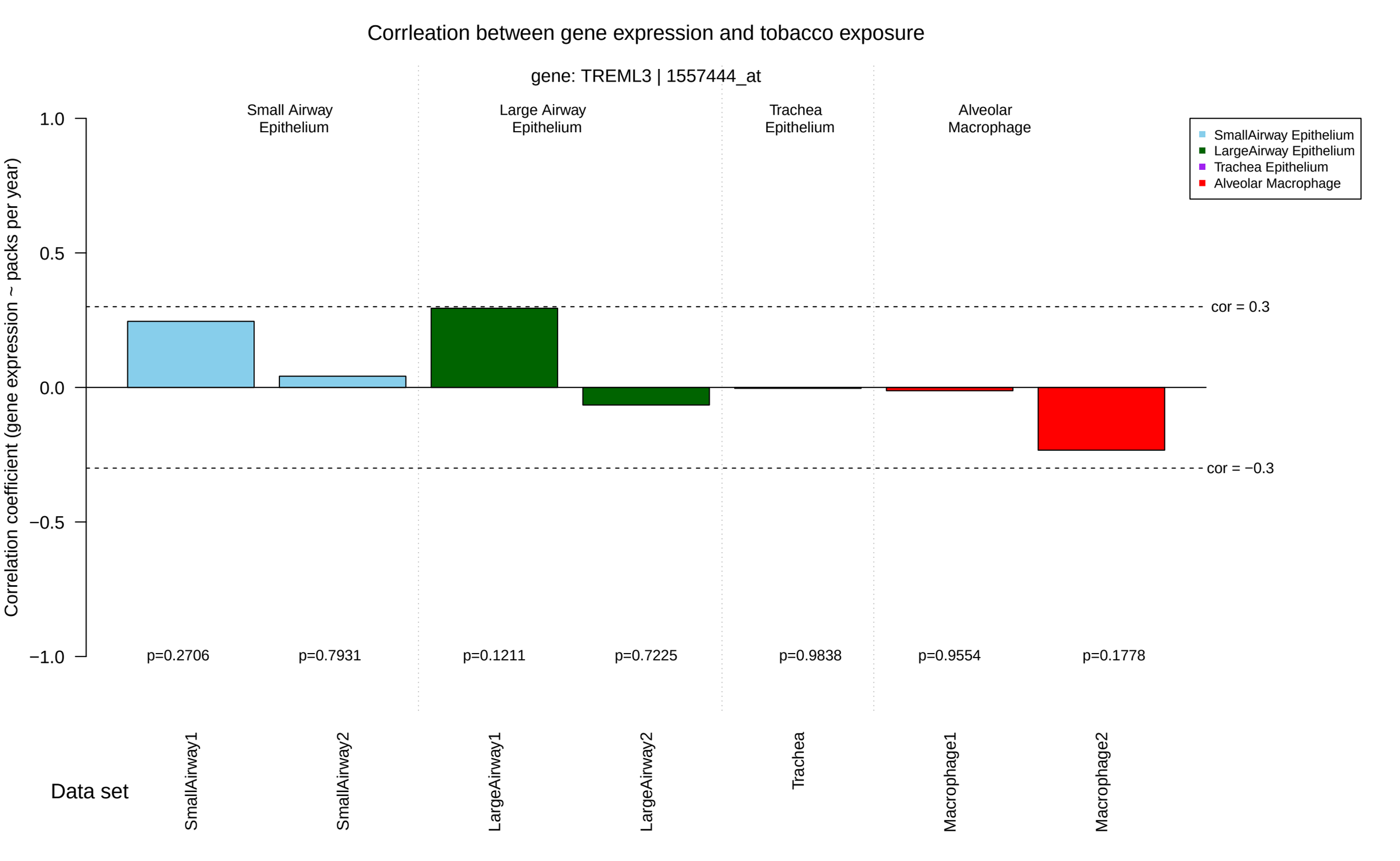

set: LargeAirway1 (p=0.0011)
TREML3 | 1557444_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3208)
TREML3 | 1557444_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7953)
TREML3 | 1557444_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5792)
TREML3 | 1557444_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.081)
TREML3 | 1557444_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TREML3 | 1557444_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2706
p=0.7931
p=0.1211
p=0.7225
p=0.9838
p=0.9554
p=0.1778
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2